

Laidlaw Stage 3 endorsement — one page

Shen Ruililin · Professor David Bishai · 24 August 2026 · not a journal manuscript

This page is only for the HKU Laidlaw research-report form. The programme essay is *ShenRuililin_Laidlaw_Stage3Report.pdf* (24 August rebuild). Journal writing continues in the live collaborative file and is a separate object. Do not sign that live file as the programme report.

What the endorsement covers

Programme file	Research report due 31 August 2026 . Poster file due 15 September (information only; no signature now).
Ask	Satisfactory / unsatisfactory rating, comments, name, signature, date. Please by Friday 28 August , or on paper at the Thursday 27 August meeting.
Estimand	Monthly CHD and HF first hospitalisation after first diagnosis, T2D/HTN cohort, Hong Kong, 2013–2023 (132 months). Ecological count ratios, not incidence.
Not claimed	Confirmatory thermal effect; principal-diagnosis CHD/HF; AMI; stroke result; daily DLNM triggering; excess-death totals from the lab's mortality paper; Model 2 or Model 3 health coefficients.

What the report already says

- Admission cause was not recorded. Stroke was named and not attached; no stroke result is reported.
- Model 1: twelve thermal comparisons, specified after the outcome series were available. After BH correction, all $q > 0.19$ (both leading estimates have $q = 0.192$).
- Model 2 (monthly humidity and rainfall) and Model 3 (days above or below the historical same-calendar-day mean) are specified and not fitted in this report.
- Exploratory CHD hot-night count ratio 1.022 (1.002–1.042) per five extra hot nights, under Newey–West lag 6; two of four uncertainty methods include 1; the association is not evident in 2013–2019.
- Exploratory HF cold-day count ratio 1.073 (1.006–1.144) per five extra cold days; concordant across standard-error methods, still q -unprotected. Official cold days fall almost entirely in December–February (141 of 145).
- The conclusion says CHD counts were more closely associated with hot nights, and HF counts with cold days, than with the other thermal measures. That ranking is not a multiplicity-protected difference, and it is not a journal headline.

Ask of you

1. Complete the supervisor block on form 2a (the student completes his own fields; yours stay blank until you write them).
2. Confirm that these disclosure-minimised aggregate results may go to laidlaw@hku.hk after any correction you request.
3. Ignore the live manuscript and the lab heatwave-mortality paper for this signature. Those are separate files.

Optional comments paste (shorten or replace): I have examined the research report. Shen Ruililin completed a clear exploratory analysis of monthly CHD and HF first hospitalisations among people with diabetes or hypertension in Hong Kong, 2013–2023. Reported estimates are from Model 1; all q -values exceed 0.19. The report does not claim a confirmatory thermal effect, notes that admission cause was not recorded and that stroke data were not delivered, and is suitable for the Laidlaw Stage 3 research-report requirement. I rate the student's performance as satisfactory.

Heat, cold and first hospitalisation after cardiovascular diagnosis in Hong Kong

An exploratory monthly study among people with diabetes or hypertension, 2013–2023

Shen Ruililin

Laidlaw Scholars Programme · The University of Hong Kong · August 2026

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Abstract

Background. Hong Kong is recording more hot nights than a decade ago, while cold days persist. People with type 2 diabetes or hypertension already face high risk of coronary heart disease (CHD) and heart failure (HF). Whether monthly temperature patterns are associated with monthly first-hospitalisation counts in this group is untested.

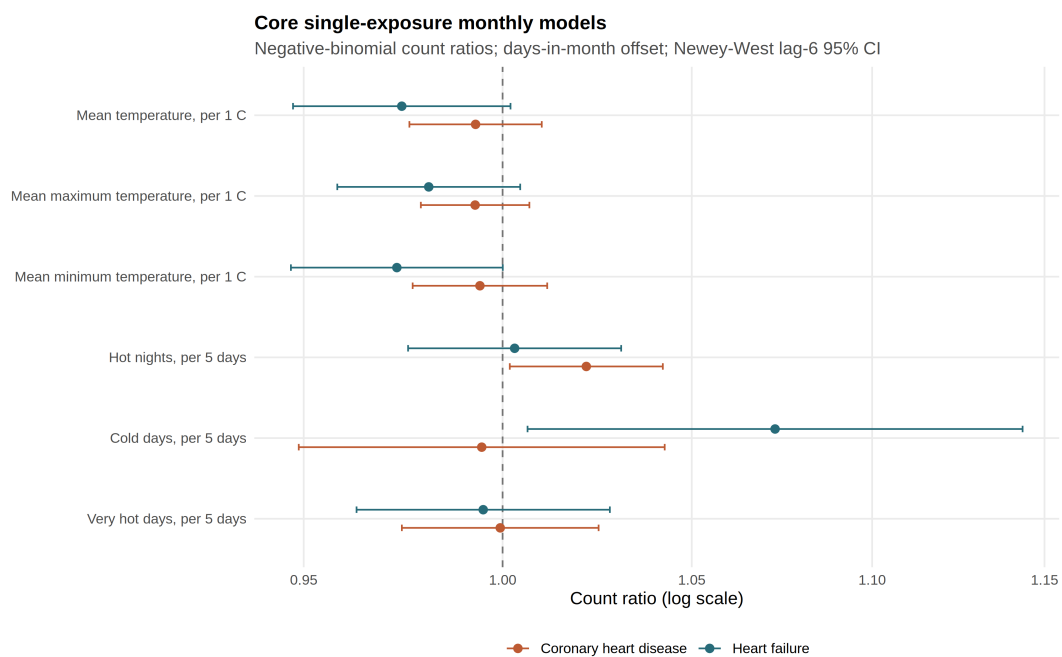
Methods. Governed Hospital Authority aggregates supplied monthly counts of first hospitalisation after a first CHD or HF diagnosis among people with type 2 diabetes and/or hypertension, January 2013–December 2023 (132 months). Admission cause was not recorded. Model 1 is a separate negative-binomial fit for each of six thermal measures and each outcome: monthly mean, mean maximum and mean minimum temperature, and official counts of hot nights, very hot days and cold days, with calendar-month indicators, a smooth long-term trend, and a days-in-month offset. Model 2 adds monthly mean relative humidity and monthly total rainfall. Model 3 replaces official extreme-day counts with days warmer or cooler than the historical same-calendar-day mean. Reported estimates are from Model 1. Intervals used model-based, HC1, and Newey–West lag-3 and lag-6 constructions. Benjamini–Hochberg q -values covered the twelve Model 1 comparisons.

Results. Under Newey–West lag 6, five additional hot nights corresponded to a CHD count ratio of 1.022 (1.002–1.042) and five additional cold days to an HF count ratio of 1.073 (1.006–1.144), both with $q = 0.192$; all twelve q -values exceeded 0.19. For CHD hot nights, model-based and HC1 intervals included 1, as did the pre-2020 estimate (1.011, 0.991–1.032). For HF cold days, all four constructions excluded 1, and the pre-2020 estimate was stronger (1.113, 1.053–1.176). Official cold days fell almost entirely in December–February (141 of 145).

Conclusions. The data do not support a multiplicity-protected thermal claim. The HF association is concordant across standard-error methods; the CHD association depends on the construction and on including 2020–2023. Estimates are ecological monthly count ratios for a first-event outcome without recorded admission cause.

Table 2. Model 1: separate negative-binomial models with a days-in-month offset and Newey–West lag-6 intervals. Thermal exposures only.

Outcome	Exposure contrast	Count ratio (95% CI)	<i>q</i>
CHD	Mean temperature / 1 °C	0.993 (0.976–1.010)	0.726
CHD	Mean maximum temperature / 1 °C	0.993 (0.979–1.007)	0.642
CHD	Mean minimum temperature / 1 °C	0.994 (0.977–1.012)	0.764
CHD	Hot nights / 5 days	1.022 (1.002–1.042)	0.192
CHD	Cold days / 5 days	0.995 (0.949–1.043)	0.900
CHD	Very hot days / 5 days	0.999 (0.974–1.025)	0.962
HF	Mean temperature / 1 °C	0.974 (0.947–1.002)	0.207
HF	Mean maximum temperature / 1 °C	0.981 (0.958–1.005)	0.272
HF	Mean minimum temperature / 1 °C	0.973 (0.947–1.000)	0.202
HF	Hot nights / 5 days	1.003 (0.976–1.031)	0.900
HF	Cold days / 5 days	1.073 (1.006–1.144)	0.192
HF	Very hot days / 5 days	0.995 (0.963–1.028)	0.900

**Figure 2.** Count ratios and 95% confidence intervals for the twelve Model 1 monthly fits. The dashed line at 1 indicates no estimated difference. All twelve Benjamini–Hochberg *q*-values exceeded 0.19.

Uncertainty and robustness

The CHD hot-night point estimate was 1.022 under all four standard-error methods. Intervals were 0.995–1.049 (model-based), 0.997–1.047 (HC1), 1.0003–1.0439 (Newey–West lag 3), and 1.002–1.042 (Newey–West lag 6). The first two include 1; the Newey–West intervals exclude it, at lag 3 only just. Residual autocorrelation at lag 1 was 0.508 in this model. The Newey–West intervals were also narrower

The study therefore offers hypotheses, not proof of thermal effects. What it leaves behind is a precisely defined first-event outcome, a complete twelve-comparison Model 1 panel, uncertainty shown under four methods, and a documented refusal to produce daily coefficients that monthly counts cannot support.

Acknowledgements

I am especially grateful to Professor David Makram Bishai, who supervised this Laidlaw Scholars project. He set its scientific direction: a complete comparison of heat and cold encodings in one Hong Kong panel rather than a search for a single confirmatory association, with every comparison reported, uncertainty shown under more than one standard-error method, and the limits of monthly aggregates stated in the same document as the estimates. He advised on population-health interpretation of monthly counts and guided what may honestly be reported from governed Hospital Authority aggregates. The study sits within a School of Public Health programme he leads on temperature and health in Hong Kong, which also includes complementary modelled heatwave-mortality work. This report is submitted for the Laidlaw Stage 3 requirement under his supervision. Journal writing continues separately with the weather and health-data co-investigators. Remaining errors are mine.

I thank Hogan for guidance on weather definitions and on academic writing, and Zhenyuan Liu for constructing and releasing the governed Hospital Authority aggregates and for regression mentorship. Diagnosis-record construction was assisted by Dr Jingjing Zhou. Dissemination of governed results requires the approval of the team and the relevant institutions.